

Appendix

1) Mathematical Model of Power Sources

The electrical power supplied to the distribution network is mainly provided by the main grid, dispatchable DG units, and wind power generation. The corresponding mathematical models are formulated as follows:

$$P_{i,t}^{\text{Grid,min}} \leq P_{i,t}^{\text{Grid}} \leq P_{i,t}^{\text{Grid,max}} \quad (\text{A1})$$

$$Q_{i,t}^{\text{Grid,min}} \leq Q_{i,t}^{\text{Grid}} \leq Q_{i,t}^{\text{Grid,max}} \quad (\text{A2})$$

$$P_i^{\text{DG,min}} u_{i,t}^{\text{DG}} \leq P_{i,t}^{\text{DG}} \leq P_i^{\text{DG,max}} u_{i,t}^{\text{DG}} \quad (\text{A3})$$

$$Q_i^{\text{DG,min}} u_{i,t}^{\text{DG}} \leq Q_{i,t}^{\text{DG}} \leq Q_i^{\text{DG,max}} u_{i,t}^{\text{DG}} \quad (\text{A4})$$

$$P_{i,t+1}^{\text{DG}} - P_{i,t}^{\text{DG}} \leq P_i^{\text{ramp}} \quad (\text{A5})$$

$$P_{i,t}^{\text{DG}} - P_{i,t+1}^{\text{DG}} \leq P_i^{\text{ramp}} \quad (\text{A6})$$

$$0 \leq P_{i,t}^{\text{WP}} \leq P_i^{\text{WP,PRE}} \quad (\text{A7})$$

$$C^{\text{Gen}} = \pi^{\text{Grid}} \sum_t P_{i,t}^{\text{Grid}} + \sum_t \sum_i c_i^{\text{DG}} P_{i,t}^{\text{DG}} \quad (\text{A8})$$

where $P_{i,t}^{\text{Grid}}$ and $Q_{i,t}^{\text{Grid}}$ denote the active and reactive power exchanged with the main grid, respectively. $P_i^{\text{Grid,min}}, P_i^{\text{Grid,max}}, Q_i^{\text{Grid,min}}$ and $Q_i^{\text{Grid,max}}$ represent the corresponding lower and upper limits of the main-grid power exchange. $P_{i,t}^{\text{DG}}$ and $Q_{i,t}^{\text{DG}}$ are the active and reactive power outputs of the dispatchable DG unit, while $P_i^{\text{DG,min}}, P_i^{\text{DG,max}}, Q_i^{\text{DG,min}}$ and $Q_i^{\text{DG,max}}$ denote its operating limits. The binary variable $u_{i,t}^{\text{DG}}$ indicates the commitment status of the dispatchable DG unit. P_i^{ramp} denotes the ramping limit. $P_{i,t}^{\text{WP}}$ represents the scheduled wind power output, and $P_i^{\text{WP,PRE}}$ denotes the forecasted available wind power. C^{Gen} denotes the total generation cost, while π^{Grid} and c_i^{DG} represent the electricity purchase price from the main grid and the DG generation cost, respectively. Equations (A1) and (A2) define the active and reactive power exchange limits between the distribution network and the main grid. Equations (A3) and (A4) impose the active and reactive power output limits of dispatchable DG units. Equations (A5) and (A6) describe the ramping constraints of DG active power output between adjacent periods. Equation (A7) ensures that the scheduled wind power output does not exceed the forecasted maximum available wind power. Equation (A8) calculates the total generation cost, including the electricity purchase cost from the main grid and the generation cost of DG units.

2) Mathematical Model of Distribution Network

The distribution network is described by LinDistFlow. The power balance equation for the distribution network is as follows:

$$P_{i,t}^{\text{Grid}} + \sum_{g \in G} P_{i,t}^{\text{DG}} + P_{i,t}^{\text{WP}} + \sum_{j \in I} P_{ji,t} = P_{i,t}^{\text{D}} + P_{i,t}^{\text{P2H}} + \sum_{k \in I} P_{ik,t} \quad (\text{A9})$$

$$Q_{i,t}^{\text{Grid}} + \sum_{g \in G} Q_{i,t}^{\text{DG}} + \sum_{j \in I} Q_{ji,t} = Q_{i,t}^{\text{D}} + \sum_{k \in I} Q_{ik,t} \quad (\text{A10})$$

$$V_{j,t} = V_{i,t} - 2(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}) \quad (\text{A11})$$

$$V_i^{\min} \leq V_{i,t} \leq V_i^{\max} \quad (\text{A12})$$

$$-P_{ij}^{\max} \leq P_{ij,t} \leq P_{ij}^{\max} \quad (\text{A13})$$

$$-Q_{ij}^{\max} \leq Q_{ij,t} \leq Q_{ij}^{\max} \quad (\text{A14})$$

where $P_{ij,t}$ and $Q_{ij,t}$ are the active and reactive power flows from bus i to j , respectively. $P_{i,t}^{\text{D}}$ and $Q_{i,t}^{\text{D}}$ denote the net active and reactive power demands. $V_{i,t}$ is the squared voltage magnitude. r_{ij} and x_{ij} are the resistance and reactance of branch ij . $V_i^{\min}$ and $V_i^{\max}$ are the lower and upper voltage limits. $P_{ij}^{\max}$ and $Q_{ij}^{\max}$ denote the active and reactive power flow limits. Equations (A9) and (A10) enforce the active and reactive power balance at each bus. Equation (A11) describes the voltage drop along each distribution branch. Equation (A12) maintains the bus voltage magnitude within its permissible range. Equations (A13) and (A14) impose the active and reactive branch power flow limits.

3) Parameters used in case studies

Parameter	Symbol	Value
Total hydrogen demand	F_t^{load}	15MW
Hydrogen storage tank capacity	$\bar{S}$	5MW
Unit degradation cost per EOH	c^{EOH}	6.25 \$
Electricity purchase price from the main grid	π^{Grid}	20 \$/MWh
Auxiliary power of each electrolyzer unit	P^a	0.05MW

4) Piecewise-linear approximation of electrolyzer efficiency

Segment	Normalized power range	Hydrogen-production coefficient
1	10%-20%	0
2	20%-40%	0.611
3	40%-60%	0.593
4	60%-80%	0.576
5	80%-100%	0.560